

Derivation of the Disturbance Equations

From the compressible Navier–Stokes equations to the
disturbance eigenvalue problem solved by `pyMack`

`pyMack` technical documentation

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1 Scope

Linear stability theory (LST) determines whether an infinitesimal wavelike disturbance superposed on a steady laminar boundary layer amplifies or decays as it propagates [1]. Because the disturbance is infinitesimal, products of disturbance quantities are negligible and its governing equations are linear; because their coefficients depend, under the parallel-flow assumption, only on the wall-normal coordinate y , the normal-mode substitution $q' = \hat{q}(y) e^{i(\alpha x - \omega t)}$ removes x and t entirely, leaving an eigenvalue problem for the shape $\hat{q}(y)$.

This document derives that eigenvalue problem: the linearisation and normal-mode ansatz (Section 2), the four resulting ordinary differential equations (Section 3), their collection into operator form and Chebyshev discretisation into the finite eigenvalue problem the solvers receive (Section 4), the physical character of the instability modes (Section 5), and the idealisations that bound the derivation (Section 6). The mean-flow profiles that enter as coefficients are taken here as known data. Figure 1 places these stages in the full computation.

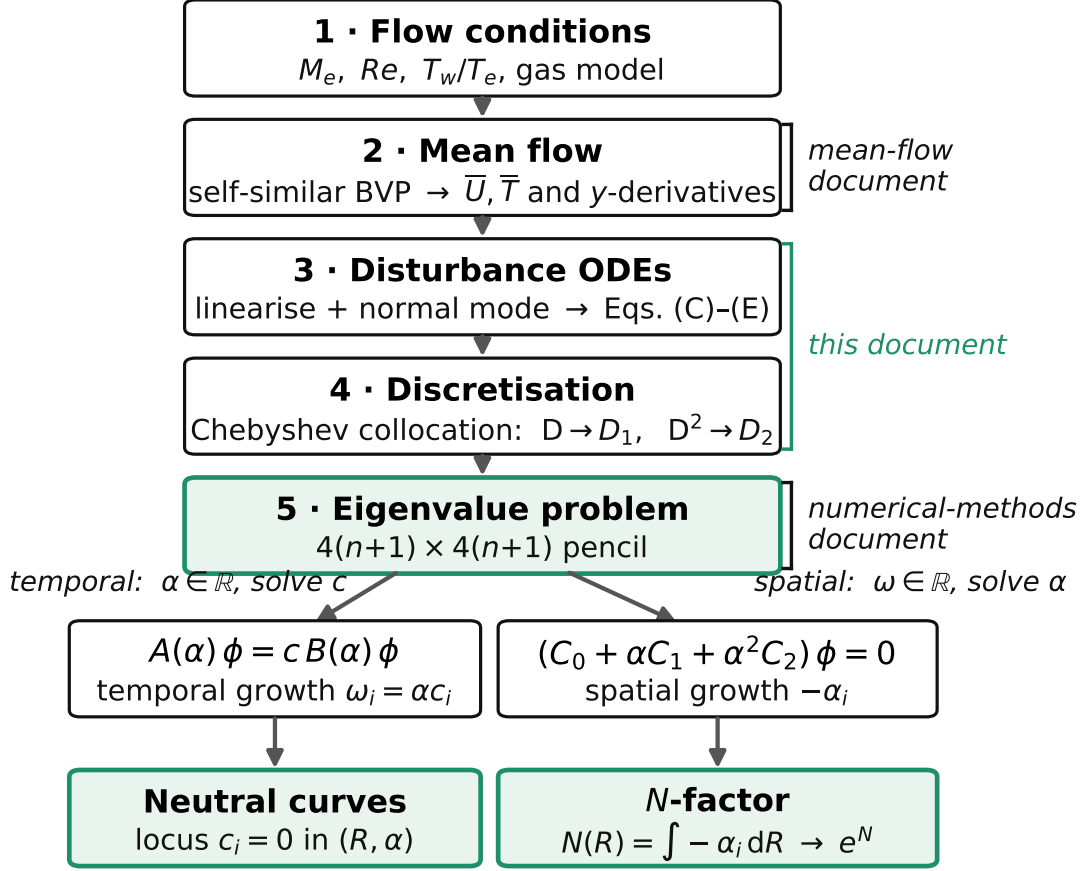


Figure 1: The complete LST computation: the base flow (stages 1–2), the disturbance equations and their discretisation derived here (stages 3–4), and the eigenvalue solution (stage 5); the two solution routes are temporal (fixed real α , complex c) and spatial (fixed real ω , complex α).

Notation. Overbars denote mean (base-flow) quantities; hats denote complex disturbance amplitudes. The continuous wall-normal amplitude vector is $\hat{q}(y) = (\hat{u}, \hat{v}, \hat{T}, \hat{p})$. A prime on a mean quantity denotes $D \equiv d/dy$; from Section 4 onward the distinction between the continuous operator D and its discrete counterpart is made explicit.

Base flow & transport		Disturbance / eigenvalue	
$\bar{U}, \bar{T}, \bar{\rho}=1/\bar{T}$	mean velocity, temp., density	$\hat{q}$	amplitude shape, Eq. (6)
$\bar{\mu}, \bar{\kappa}$	viscosity, conductivity	$\hat{u}, \hat{v}, \hat{T}, \hat{p}$	velocity/temp./pressure amplitudes
M_e, Re, Pr, γ	Mach, Reynolds, Prandtl, c_p/c_v	α, ω	wavenumber, frequency
$\nu_R=1/(Re\bar{\rho})$	viscous prefactor	$c=\omega/\alpha$	complex phase speed; c_r, c_i its parts
$\mathcal{C}_\kappa, \mathcal{C}_d, \mathcal{L}_c$	conduction/dissipation groups	M_{rel}	relative Mach number, §5

2 Linearisation and the normal-mode ansatz

The flow is governed by the compressible Navier–Stokes equations, expressing conservation of mass, momentum, and energy for a viscous, heat-conducting perfect gas [2]:

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{u} = 0, \quad (\text{mass}) \quad (1)$$

$$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \nabla \cdot \boldsymbol{\tau}, \quad (\text{momentum}) \quad (2)$$

$$\rho c_p \frac{DT}{Dt} = \frac{Dp}{Dt} + \nabla \cdot (\kappa \nabla T) + \Phi, \quad (\text{energy}) \quad (3)$$

$$p = \rho RT, \quad (\text{state}) \quad (4)$$

where $\mathbf{u} = (u, v)$ is the velocity, ρ the density, T the temperature, p the pressure, μ the viscosity, κ the conductivity, c_p the specific heat, R the gas constant, $D/Dt = \partial_t + \mathbf{u} \cdot \nabla$ the material derivative, $\boldsymbol{\tau}$ the Newtonian viscous stress tensor, and $\Phi = \boldsymbol{\tau} : \nabla \mathbf{u}$ the dissipation function (both linear in the velocity gradients). These are the same compressible Navier–Stokes equations solved for the steady mean flow; here they are applied a second time, in linearised form, for the disturbance.

Step 1: decomposition into mean and disturbance. Each field is split as

$$u = \bar{U}(y) + u', \quad v = v', \quad T = \bar{T}(y) + T', \quad p = \bar{p} + p', \quad \rho = \bar{\rho}(y) + \rho', \quad |\cdot'| \ll |\cdot|, \quad (5)$$

with $\bar{v} = 0$ for a parallel mean flow. The collection $q' = (u', v', T', p', \rho')$ is the disturbance.

Step 2: linearisation. Substituting (5) into (1)–(4), the mean satisfies the equations by itself and cancels; terms quadratic in the primed quantities are dropped. The remaining equations are linear in q' with coefficients constructed from the known mean profiles.

Step 3: parallel-flow assumption. The boundary layer grows slowly, so locally $\bar{q} = \bar{q}(y)$ only and the coefficients depend on y alone. This neglects the $O(1/R)$ streamwise growth of the mean flow and of the eigenfunction, which the parabolised stability equations retain [10] (cf. Section 6).

Step 4: normal-mode ansatz. The disturbance is represented as a single travelling wave,

$$q'(x, y, t) = \hat{q}(y) e^{i(\alpha x - \omega t)}, \quad \hat{q} = (\hat{u}, \hat{v}, \hat{T}, \hat{p})^\top, \quad (6)$$

with α the streamwise wavenumber, ω the frequency, and $c = \omega/\alpha$ the complex phase speed. Substitution replaces $\partial_x \mapsto i\alpha$, $\partial_t \mapsto -i\omega$, and $\partial_y \mapsto D$, collapsing the linear PDEs to coupled ODEs in y (Figure 2). The spanwise wavenumber is set to zero ($\beta = 0$); a general oblique wave $\propto e^{i(\alpha x + \beta z - \omega t)}$ propagates at angle $\psi = \arctan(\beta/\alpha)$, and since Squire’s theorem does not carry over to compressible flow, the first mode is most unstable oblique ($\psi \sim 55^\circ$) while the second mode is most unstable at $\psi = 0$ [1]. The consequences of the two-dimensional restriction are stated in Section 6.

Step 5: elimination of density. The linearised equation of state gives

$$\frac{\rho'}{\bar{\rho}} = \gamma M_e^2 \hat{p} - \frac{\hat{T}}{\bar{T}}, \quad (7)$$

which removes ρ' from the system. Four amplitude fields remain, $\hat{u}, \hat{v}, \hat{T}, \hat{p}$, with $\hat{p}$ scaled by $\rho_e U_e^2$ (so that $\gamma M_e^2 \hat{p} = p'/p_e$).

A worked linearisation. To make the mechanics explicit, consider mass conservation (1), written out as $\partial_t \rho + \partial_x(\rho u) + \partial_y(\rho v) = 0$. Inserting the decomposition (5) and expanding every product sorts the equation into three groups:

$$\underbrace{\partial_t \bar{\rho} + \partial_x(\bar{\rho} \bar{U})}_{\text{mean: vanishes}} + \underbrace{\partial_t \rho' + \bar{U} \partial_x \rho' + \bar{\rho} \partial_x u' + \partial_y(\bar{\rho} v')}_{\text{linear in } q': \text{ retained}} + \underbrace{\partial_x(\rho' u') + \partial_y(\rho' v')}_{O(|q'|^2): \text{ dropped}} = 0. \quad (8)$$

The mean group vanishes identically: the base flow is steady, $\partial_t \bar{\rho} = 0$, and parallel, so $\bar{\rho}$ and $\bar{U}$ depend on y alone and $\partial_x(\bar{\rho} \bar{U}) = 0$; in general the mean satisfies its own conservation equations and cancels from the disturbance system. The same parallel-flow property has already simplified the linear group, since $\partial_x(\bar{\rho} u') = \bar{\rho} \partial_x u'$ and $\partial_x(\rho' \bar{U}) = \bar{U} \partial_x \rho'$. The quadratic group is negligible by the smallness assumption in (5). What survives is the linear equation

$$\partial_t \rho' + \bar{U} \partial_x \rho' + \bar{\rho} \partial_x u' + \partial_y(\bar{\rho} v') = 0. \quad (9)$$

Applying the ansatz (6) replaces $\partial_t \rightarrow -i\alpha c$, $\partial_x \rightarrow i\alpha$, $\partial_y \rightarrow D$ and each primed field by its amplitude, so that the first two terms combine into the convective factor:

$$i\alpha(\bar{U} - c) \hat{\rho} + i\alpha \bar{\rho} \hat{u} + \bar{\rho} D \hat{v} + \bar{\rho}' \hat{v} = 0. \quad (10)$$

Dividing by $\bar{\rho}$, substituting $\hat{\rho}/\bar{\rho} = \gamma M_e^2 \hat{p} - \hat{T}/\bar{T}$ from (7), and using $\bar{\rho}'/\bar{\rho} = -\bar{T}'/\bar{T}$ (from $\bar{\rho} = 1/\bar{T}$) yields exactly Eq. (11) below. The remaining three equations follow by the identical procedure applied to (2)–(3); the algebra is mechanical but lengthy, and only the result is printed.

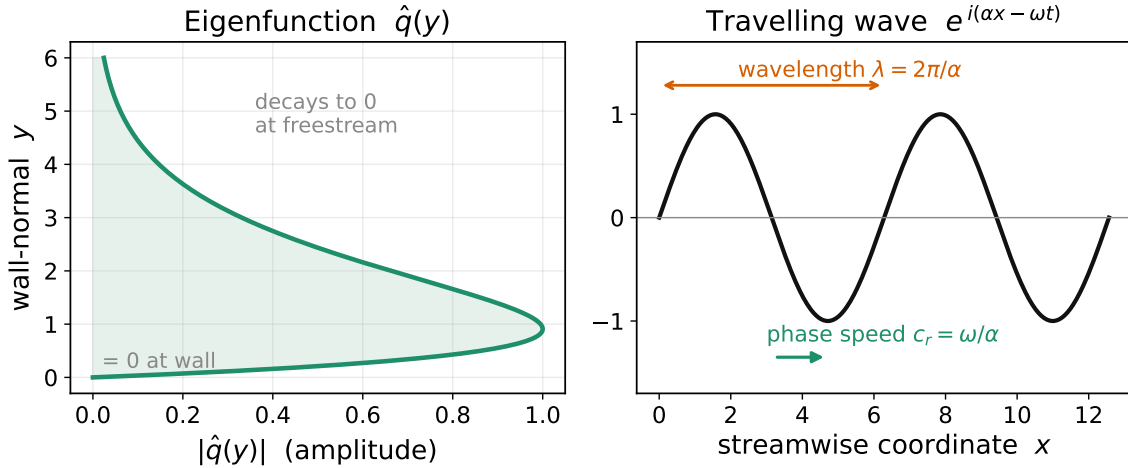


Figure 2: The ansatz (6) factorises a disturbance $q' = \hat{q}(y) e^{i(\alpha x - \omega t)}$ into a wall-normal *shape* $\hat{q}(y)$ (left; the eigenfunction, zero at the wall and decaying at the edge) and a travelling wave in x (right; wavelength $2\pi/\alpha$, phase speed $c_r = \text{Re}(c)$).

3 The four disturbance equations

The linearised system consists of four coupled ODEs in y for $\hat{u}, \hat{v}, \hat{T}, \hat{p}$: continuity (with the equation of state folded in through (7)), streamwise and wall-normal momentum, and energy. Fixing one of α or ω real leaves the other (through c or α) as the eigenvalue. Every overbarred quantity is a known base-flow coefficient; all transport derivatives ($d\mu/dT$, $d\kappa/dT$, ...) are evaluated at the local mean temperature $\bar{T}(y)$ and are therefore known functions of y . The form below follows Özgen

& Kircali [3]; it is equivalent to the canonical compressible stability equations of Mack [1] and Malik [4] restricted to two dimensions. Momentum and energy are divided by $\bar{\rho}$, and the viscous prefactor is written $\nu_R \equiv 1/(Re \bar{\rho})$; every $\bar{\rho}$ is the base-flow density $\bar{\rho}(y) = 1/\bar{T}(y)$, uniformly inside ν_R , $\mathcal{C}_\kappa$, and $\mathcal{C}_d$.

(C) Continuity + equation of state.

$$i\alpha\hat{u} + D\hat{v} - \frac{\bar{T}'}{\bar{T}}\hat{v} + i\alpha(\bar{U} - c)\left(\gamma M_e^2\hat{p} - \frac{\hat{T}}{\bar{T}}\right) = 0. \quad (11)$$

(X) Streamwise momentum.

$$i\alpha(\bar{U} - c)\hat{u} + \bar{U}'\hat{v} + i\alpha\bar{\rho}^{-1}\hat{p} = \nu_R\left[\bar{\mu}(D^2 - \frac{4}{3}\alpha^2)\hat{u} + \bar{\mu}'D\hat{u} + i\alpha(\frac{1}{3}\bar{\mu}D + \bar{\mu}')\hat{v} + \frac{d\bar{\mu}}{dT}(\bar{U}'' + \bar{U}'D)\hat{T} + \frac{d^2\bar{\mu}}{dT^2}\bar{U}'\bar{T}'\hat{T}\right]. \quad (12)$$

(Y) Wall-normal momentum.

$$i\alpha(\bar{U} - c)\hat{v} + \bar{\rho}^{-1}D\hat{p} = \nu_R\left[\bar{\mu}(\frac{4}{3}D^2 - \alpha^2)\hat{v} + \frac{4}{3}\bar{\mu}'D\hat{v} + i\alpha(\frac{1}{3}\bar{\mu}D - \frac{2}{3}\bar{\mu}')\hat{u} + i\alpha\frac{d\bar{\mu}}{dT}\bar{U}'\hat{T}\right]. \quad (13)$$

(E) Energy (temperature).

$$i\alpha(\bar{U} - c)\hat{T} + \bar{T}'\hat{v} + (\gamma - 1)\bar{\rho}^{-1}(i\alpha\hat{u} + D\hat{v}) = \mathcal{C}_\kappa\mathcal{L}_c\hat{T} + \mathcal{C}_d\left[2\bar{\mu}\bar{U}'(D\hat{u} + i\alpha\hat{v}) + \frac{d\bar{\mu}}{dT}(\bar{U}')^2\hat{T}\right], \quad (14)$$

$$\mathcal{L}_c = D^2 - \alpha^2 + \frac{1}{\bar{\kappa}}\frac{d\bar{\kappa}}{dT}(2\bar{T}'D + \bar{T}'') + \frac{1}{\bar{\kappa}}\frac{d^2\bar{\kappa}}{dT^2}(\bar{T}')^2, \quad \mathcal{C}_\kappa = \frac{\gamma\bar{\mu}}{Pr Re \bar{\rho}}, \quad \mathcal{C}_d = \frac{\gamma(\gamma - 1)M_e^2}{Re \bar{\rho}}. \quad (15)$$

The Prandtl number in $\mathcal{C}_\kappa$ is taken as the local value when the selected transport model supplies a temperature-dependent Pr ; for the constant- Pr default the two coincide.

Structure of the equations. The following observations locate the physics in (11)–(14):

- $i\alpha(\bar{U} - c)$ is convection by the mean flow relative to the wave; it vanishes at the *critical layer* $\bar{U} = c_r$, where the inviscid form of the equations becomes singular, which is why that layer governs much of the instability physics [1].
- D, D^2 are wall-normal derivatives; the $-\alpha^2$ terms arise from $\partial_x^2 \mapsto -\alpha^2$.
- The $\nu_R = 1/(Re\bar{\rho})$ terms are viscous; the factors $\frac{4}{3}, \frac{1}{3}$ correspond to the Stokes hypothesis for the bulk viscosity.

4 Operator form and the discretised derivative

Equations (11)–(14) are linear both in the amplitude vector $\hat{q} = (\hat{u}, \hat{v}, \hat{T}, \hat{p})^\top$ and in the eigenvalue, and every derivative they contain is of at most second order. They can therefore be collected into the single operator statement

$$\mathbf{L}(\alpha; y, D) \hat{q} = c \mathbf{M}(\alpha; y) \hat{q}, \quad \mathbf{L} = \mathbf{A}_2(y) D^2 + \mathbf{A}_1(\alpha; y) D + \mathbf{A}_0(\alpha; y), \quad (16)$$

with rows ordered (C, X, Y, E) and columns ordered $(\hat{u}, \hat{v}, \hat{T}, \hat{p})$. Two structural facts make (16) particularly convenient for what follows.

First, the eigenvalue c enters (11)–(14) *only* through the convective factor $i\alpha(\bar{U} - c)$. The right-hand operator $\mathbf{M}$ is therefore purely algebraic, containing no derivatives, and can be written down at once:

$$\mathbf{M}(\alpha; y) = i\alpha \begin{pmatrix} 0 & 0 & -1/\bar{T} & \gamma M_e^2 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (17)$$

Second, only three terms in the entire system carry D^2 : the viscous diffusion of $\hat{u}$ and $\hat{v}$ in (12)–(13) and the heat conduction of $\hat{T}$ in (14). The second-order coefficient matrix is therefore sparse:

$$\mathbf{A}_2(y) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ -\nu_R \bar{\mu} & 0 & 0 & 0 \\ 0 & -\frac{4}{3} \nu_R \bar{\mu} & 0 & 0 \\ 0 & 0 & -\mathcal{C}_\kappa & 0 \end{pmatrix}, \quad (18)$$

where the signs follow from moving the right-hand sides of (12)–(14) to the left. The pressure amplitude $\hat{p}$ is never differentiated twice, so its column in $\mathbf{A}_2$ is empty, reflecting the absence of a pressure diffusion mechanism.

The first- and zeroth-order matrices are read off from (11)–(14) in the same manner. For the energy row it is convenient to split the conduction operator (15) by derivative order,

$$\mathcal{L}_c = D^2 + \ell_1 D + \ell_0, \quad \ell_1 = \frac{2\bar{T}'}{\bar{\kappa}} \frac{d\kappa}{dT}, \quad \ell_0 = -\alpha^2 + \frac{\bar{T}''}{\bar{\kappa}} \frac{d\kappa}{dT} + \frac{(\bar{T}')^2}{\bar{\kappa}} \frac{d^2\kappa}{dT^2}. \quad (19)$$

The first-order matrix collects every term carrying a single D :

$$\mathbf{A}_1(\alpha; y) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -\nu_R \bar{\mu}' & -\frac{i\alpha}{3} \nu_R \bar{\mu} & -\nu_R \frac{d\mu}{dT} \bar{U}' & 0 \\ -\frac{i\alpha}{3} \nu_R \bar{\mu} & -\frac{4}{3} \nu_R \bar{\mu}' & 0 & \bar{\rho}^{-1} \\ -2\mathcal{C}_d \bar{\mu} \bar{U}' & (\gamma - 1) \bar{\rho}^{-1} & -\mathcal{C}_\kappa \ell_1 & 0 \end{pmatrix}, \quad (20)$$

and the zeroth-order matrix collects everything algebraic that does not multiply c :

$$\mathbf{A}_0(\alpha; y) = \begin{pmatrix} i\alpha & -\frac{\bar{T}'}{\bar{T}} & -\frac{i\alpha \bar{U}}{\bar{T}} & i\alpha \gamma M_e^2 \bar{U} \\ i\alpha \bar{U} + \frac{4}{3} \alpha^2 \nu_R \bar{\mu} & \bar{U}' - i\alpha \nu_R \bar{\mu}' & -\nu_R \left(\frac{d\mu}{dT} \bar{U}'' + \frac{d^2\mu}{dT^2} \bar{U}' \bar{T}' \right) & i\alpha \bar{\rho}^{-1} \\ \frac{2i\alpha}{3} \nu_R \bar{\mu}' & i\alpha \bar{U} + \alpha^2 \nu_R \bar{\mu} & -i\alpha \nu_R \frac{d\mu}{dT} \bar{U}' & 0 \\ i\alpha(\gamma - 1) \bar{\rho}^{-1} & \bar{T}' - 2i\alpha \mathcal{C}_d \bar{\mu} \bar{U}' & i\alpha \bar{U} - \mathcal{C}_\kappa \ell_0 - \mathcal{C}_d \frac{d\mu}{dT} (\bar{U}')^2 & 0 \end{pmatrix}. \quad (21)$$

Temporal and spatial problems. Equation (16) supports the two standard closures. Fixing $\alpha \in \mathbb{R}$ leaves c as the eigenvalue of a *linear* operator pencil; this is the temporal problem. Fixing $\omega \in \mathbb{R}$ instead and regrouping the same equations by powers of α gives

$$(\mathbf{C}_0(\omega; y, D) + \alpha \mathbf{C}_1(\omega; y, D) + \alpha^2 \mathbf{C}_2(y)) \hat{q} = 0, \quad (22)$$

a *quadratic* pencil in α ; this is the spatial problem, reduced to linear form by companion linearisation at the numerical stage.

Chebyshev collocation. Up to this point D has denoted the continuous derivative d/dy acting on smooth functions. To make (16) computable, each amplitude is represented by its values at a finite set of grid points and D by a matrix acting on those values. For smooth functions a polynomial interpolant through Chebyshev points converges exponentially with the polynomial degree [5, 6], so few points yield many digits; this makes spectral collocation the natural discretisation for boundary-layer eigenvalue problems [4]. The Chebyshev–Gauss–Lobatto nodes on $[-1, 1]$ are

$$\xi_j = \cos\left(\frac{\pi j}{n}\right), \quad j = 0, 1, \dots, n, \quad (23)$$

clustering near the ends $\xi = \pm 1$; the ordering runs from $\xi_0 = +1$ down to $\xi_n = -1$.

The differentiation matrix. Exactly one polynomial of degree $\leq n$ passes through the sampled values $\{f(\xi_j)\}$; differentiating that interpolant and evaluating the result at the nodes is a linear operation on the values, $f'(\xi_j) = \sum_k (D_\xi)_{jk} f(\xi_k)$, with entries [7, 6]

$$(D_\xi)_{jk} = \frac{c_j}{c_k} \frac{(-1)^{j+k}}{\xi_j - \xi_k} \quad (j \neq k), \quad (D_\xi)_{jj} = -\sum_{k \neq j} (D_\xi)_{jk}, \quad c_0 = c_n = 2, \quad c_j = 1 \text{ otherwise.} \quad (24)$$

The diagonal is fixed by requiring each row to sum to zero, since a constant function must have zero derivative, and $c_0 = c_n = 2$ accounts for the doubled weight of the two endpoint nodes. As an illustration, the smallest useful case $n = 2$ (three nodes $\xi = (1, 0, -1)$) gives

$$D_\xi = \begin{pmatrix} \frac{3}{2} & -2 & \frac{1}{2} \\ \frac{1}{2} & 0 & -\frac{1}{2} \\ -\frac{1}{2} & 2 & -\frac{3}{2} \end{pmatrix}, \quad D_\xi^2 = D_\xi D_\xi = \begin{pmatrix} 1 & -2 & 1 \\ 1 & -2 & 1 \\ 1 & -2 & 1 \end{pmatrix}. \quad (25)$$

The middle row of D_ξ is the familiar central difference $\frac{1}{2}(f_0 - f_2)$, and the rows of D_ξ^2 are identical because the second derivative of the parabola through three points is constant. Both matrices are exact on polynomials of degree $\leq n$. Production runs use far more nodes, typically $n \sim 100$ – 200 , so that the eigenfunctions are resolved to several digits. At any order the matrices are dense, because every node's derivative uses every node's value, and this density is inherited by the assembled stability matrices below.

Mapping to the physical layer. The flow occupies $y \in [0, y_{\max}]$ with resolution needed at the wall, where the eigenfunctions vary fastest. The algebraic map of Malik [4],

$$y(\xi) = L \frac{1 + \xi}{1 - \xi + 2L/y_{\max}}, \quad L = \frac{y_{\max}}{6}, \quad (26)$$

sends $\xi = +1$ (node $j=0$) to $y = y_{\max}$, the freestream, and $\xi = -1$ (node $j=n$) to $y = 0$, the wall, placing roughly half the nodes below $y \approx L$. By the chain rule, with $\xi_y = 1/y_\xi$ and $\xi_{yy} = -y_\xi \xi / y_\xi^3$, the differentiation matrices on the physical grid $\{y_j\}_{j=0}^n$ are

$$D_1 = \text{diag}(\xi_y) D_\xi, \quad D_2 = \text{diag}(\xi_y^2) D_\xi^2 + \text{diag}(\xi_{yy}) D_\xi, \quad (27)$$

where the chain rule acts on the computational square D_ξ^2 : the physical D_2 is *not* $D_1 D_1$.

The discrete eigenvalue problem. Numerical solution now amounts to the substitution

$$D \longrightarrow D_1, \quad D^2 \longrightarrow D_2, \quad D_1, D_2 \in \mathbb{R}^{(n+1) \times (n+1)}. \quad (28)$$

Under (28) each mean-flow coefficient becomes a diagonal matrix, each entry of A_2, A_1, A_0, M becomes an $(n+1) \times (n+1)$ block, and the four sampled amplitudes are stacked into a single vector $\phi = (\hat{u}; \hat{v}; \hat{T}; \hat{p}) \in \mathbb{C}^{4(n+1)}$. The operator statement (16) then becomes the finite-dimensional generalised eigenvalue problem

$$\boxed{A(\alpha) \phi = c B(\alpha) \phi} \quad A, B \in \mathbb{C}^{4(n+1) \times 4(n+1)}. \quad (29)$$

Anatomy of the discrete problem. Equation (29) is the computational atom of the framework: every stability result comes from assembling and solving it, once per point of parameter space. Its three symbols carry definite meanings from the derivation.

- $A(\alpha)$ discretises the left-hand operator L of (16) and holds *all of the eigenvalue-independent physics*: convection by the mean flow, the pressure gradient, production by the mean shear, viscous diffusion, heat conduction, and the dilatational coupling, each at the chosen α .
- $B(\alpha)$ discretises the algebraic operator M of (17) and marks *where the eigenvalue enters*: since $-\mathrm{i}\omega = -\mathrm{i}\alpha c$, these are the positions where ∂_t acted before the ansatz (6).
- $\phi = (\hat{u}; \hat{v}; \hat{T}; \hat{p})$ is the eigenfunction $\hat{q}(y)$ of (16), sampled at the collocation points.

Both matrices inherit the 4×4 block structure of the operator form. With i, j over the rows (C, X, Y, E) and columns $(\hat{u}, \hat{v}, \hat{T}, \hat{p})$, every block follows one assembly rule,

$$A_{ij}(\alpha) = \Lambda([A_2]_{ij}) D_2 + \Lambda([A_1]_{ij}) D_1 + \Lambda([A_0]_{ij}), \quad B_{ij}(\alpha) = \Lambda([M]_{ij}), \quad (30)$$

where $\Lambda(f) = \text{diag}(f(y_0), \dots, f(y_n))$ places a coefficient on the grid. Three ingredients therefore build the whole problem: the differentiation matrices D_1, D_2 , diagonal samples of the mean-flow profiles, and the scalars α, M_e, Re .

One solve: input and output. A single solve of (29) takes one point of parameter space, the mean-flow profiles at (M_e, Re) and one real wavenumber α , and returns the full discrete spectrum of $4(n+1)$ eigenpairs (c, ϕ) , each a candidate normal mode:

- c_r is the phase speed as a fraction of U_e ;
- c_i sets the temporal growth rate $\omega_i = \alpha c_i$: the wave amplifies for $c_i > 0$, decays for $c_i < 0$, and is neutral at $c_i = 0$;
- the four segments of ϕ are the wave's velocity, temperature, and pressure amplitude profiles across the layer (a wall-peaked $|\hat{p}|$, for instance, signals a second mode).

Only a few of these eigenpairs are physical *discrete* modes; the rest approximate the continuous spectra or are numerical artifacts. The boundary-condition rows carry no eigenvalue and leave zero rows in B , so B is singular and the pencil is solved in generalised (QZ) form, which produces spurious infinite eigenvalues that are discarded.

Everything the framework produces is built from repeated solves of (29): one solve interrogates one point (M_e, Re, α) ; a sweep traces the growth rate across the (R, α) plane; the zero level set $c_i = 0$ of that sweep is the neutral curve; and the integral of the growth rate along frequency lines is the N -factor. Solving the pencil and identifying the physical modes within the spectrum is the remaining numerical task.

5 Physical character of the instability modes

The disturbance equations admit discrete eigenmodes; a temporal solve returns a complex phase speed such as $c = 0.930 + 0.0200i$. This section summarises the physics behind the standard mode labels, so that such a number identifies a definite wave.

First (Tollmien–Schlichting) mode. At low speed the first mode is a viscous instability ($c_r \approx 0.3$ – 0.6): viscosity introduces a phase lag between u' and v' near the wall, so the Reynolds stress $-\bar{\rho}u'v'$ no longer averages to zero and extracts energy from the mean shear. As the Mach number rises, compressibility creates a generalised inflection point (below) and the first mode becomes increasingly inviscid and inflectional, with viscosity then often acting to stabilise; it is the dominant instability only up to $M_e \approx 4$ [1].

Second (Mack) mode. Above $M_e \approx 4$ a second, usually stronger, instability appears: an acoustic wave trapped between the wall and the relative sonic line, reflecting within a near-wall acoustic waveguide [1, 13]. This interpretation accounts for its principal properties:

- its phase speed is $c_r \approx 1 - 1/M_e$, the limiting value at which the wave is sonic relative to the edge flow, which is why the spatial-solve shift target $\alpha_0 = \omega/(1 - 1/M_e)$ is effective; the realised c_r sits somewhat above this and varies along the branch (0.93 against $1 - 1/6 = 0.83$ for a representative $M_e = 6$ mode);
- it is frequency-tuned: as in any resonator the wavelength is set by the cavity thickness, so the most-amplified wavelength is $\sim 2\delta$ and the frequency scales as $f \sim U_e/2\delta$, rising as the layer thins downstream;
- its eigenfunction has a wall-peaked pressure amplitude, the acoustic field of the waveguide.

The organising quantity: relative Mach number. Both mechanisms are organised by the relative Mach number

$$M_{\text{rel}}(y) = \frac{\bar{U}(y) - c_r}{\bar{a}(y)}, \quad \bar{a} = \frac{\sqrt{T}}{M_e}, \quad (31)$$

with $\bar{a}$ the local sound speed. The critical layer is the height at which $M_{\text{rel}} = 0$, i.e. $\bar{U} = c_r$, where the eigenfunction typically peaks and the convective factor $i\alpha(\bar{U} - c)$ in (11)–(14) vanishes; the relative sonic line is $|M_{\text{rel}}| = 1$; and the second-mode acoustic waveguide is the near-wall region in which $|M_{\text{rel}}| > 1$.

The generalised inflection point. By Rayleigh’s theorem an inviscid incompressible shear flow requires a velocity inflection point ($\bar{U}'' = 0$) for instability. Lees & Lin [8] generalised this to compressible flow: the necessary condition for the inviscid first mode is a *generalised inflection point*, $(\bar{\rho}\bar{U})' = 0$. Aerodynamic heating places a hot, low-density layer near the wall and pushes that point up into the layer, giving compressible boundary layers a robust inviscid first-mode instability that low-speed layers lack. The second mode requires no inflection point; its existence condition is instead the relative-supersonic region $|M_{\text{rel}}| > 1$ above.

The wall-cooling reversal. Cooling the wall stabilises the first mode (it weakens the near-wall viscous mechanism) but destabilises the second (it thickens the relative-supersonic region and raises the tuned frequency) [1]. Cold-walled hypersonic vehicles are therefore second-mode dominated.

Phase and group velocity. c_r is the phase speed; energy propagates at the group velocity $c_g = d\omega_r/d\alpha_r$, and it is c_g that rigorously links temporal to spatial growth [9], the relation invoked whenever results are converted between the two formulations.

The following comparison places any computed eigenvalue at a glance:

	First (TS) mode	Second (Mack) mode
mechanism	viscous (low speed) $\rightarrow$ inviscid inflectional (high speed)	trapped acoustic wave (wall-sonic line)
phase speed c_r	0.3–0.6	$\approx 1 - 1/M_e$
dominant when	low speed, $M_e \lesssim 4$	high speed, $M_e \gtrsim 4$
frequency	broadband	tuned, $f \sim U_e/2\delta$
existence criterion	generalised inflection point [8]	relative-supersonic region, $ M_{\text{rel}} > 1$
wall cooling	stabilises	destabilises
most-unstable wave	oblique (3D)	2D ($\psi = 0$)

6 Assumptions and range of validity

The equations derived above are the two-dimensional, parallel-flow, modal disturbance equations. Four idealisations bound their interpretation:

1. **Parallel flow.** Setting $\bar{v} = 0$ with y -only coefficients (Step 3) neglects the $O(1/R)$ stream-wise growth of the layer, which the parabolised stability equations retain by marching the disturbance downstream [10].
2. **Two-dimensionality.** The ansatz takes $\beta = 0$ (Step 4). The most-unstable first mode is oblique ($\psi \sim 55^\circ$; Squire’s theorem does not hold under compressibility [1]), so two-dimensional first-mode growth is a lower bound; the second mode is most unstable at $\psi = 0$, so the two-dimensional result is its physical maximum.
3. **Modal analysis.** Eigenvalues describe asymptotic exponential growth; the non-normal operator also permits transient (algebraic) growth that modal analysis misses, relevant to bypass and blunt-body transition paths [11].
4. **Receptivity and e^N .** LST provides the amplification *ratio* $A(R)/A(R_0) = e^N$, not the initial amplitude $A(R_0)$; determining that amplitude is the separate receptivity problem, and the e^N method is an environment-dependent correlation rather than a first-principles criterion [12, 13].

None of these affect the equations themselves; they delimit their range of validity. A strong second mode is captured faithfully by the two-dimensional modal equations derived here, which is why they are the workhorse of hypersonic transition prediction [1, 13]. With the system (11)–(14) in operator form (16) and assembled into its discrete counterpart (29), the derivation is complete: what remains is to solve the matrix problem and identify its physical modes.

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